Smaran Teja

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Education

Northeastern University: Khoury College of Computer Sciences, Boston, MA *Sep 2023 – Present*

Candidate for Bachelor of Science in Computer Science, Expected May 2027

GPA: 3.8/4.0 (Dean's list, multiple semesters)

Relevant Courses:

Software engineering | **Distributed systems** | **Programming languages** | *Accelerated* Fundamentals of Computer Science 1 & 2 | Algorithms and Data | Object Oriented Design | Logic and Computation | Computer Systems | *Intensive* Math Reasoning

Technical Knowledge

Languages: **Java** | **Python** | C/C++ | JavaScript/Typescript | **Racket** | OCaml | HTML/CSS

Applications/software: **VS Code** | Android Studio | **Git** | Eclipse | Firebase | NetBeans | MySQL | IntelliJ IDEA | React | Node.js | Jinja | Flask | DrRacket

Experience

Research assistant: *Roulette* Probabilistic Profiler, Northeastern PRL *Sep 2025 – Present*

- Designed a novel multi-threaded profiling algorithm to profile Roulette Probabilistic Programming Language (PPL)
- Developed robust Racket algorithm to generate profiling results and HTML visualization tool in Python
- Optimized various performance and memory metrics to increase efficiency of profiler
- Evaluated multiple new heuristics for identifying possible areas of performance issues in PPLs
- Presented findings and ongoing research report at the NEU Probabilistic Programming Lab
- Currently working towards a research publication with Professor Steven Holtzen and post-doc Cameron Moy

Course development assistant: CS2100, Khoury College of Computer Sciences *May – Sep 2025*

- Led the development of course materials and infrastructure for new intro CS course
- Built Python autograders via GitHub Actions with support for mutation testing and HTML report generation
- Implemented dependency caching algorithm across runs that reduced runtime by about **75%**
- Onboarded new professors to the grading interface and autograder infrastructure

Projects and other roles

Teaching Assistant: Logic and Computation, Northeastern University *Jan 2026 – Present*

- Held midterm review session with 30+ students clarifying doubts and reviewing a practice exam
- Co-developed Ocaml library for developing test suites to run student code against and provide immediate results

Teaching Assistant: Fundamentals 1, Northeastern University *Sep – Dec 2024*

- Lead developer and maintainer for ***FeedBot*** tool deployed for the first time, now used campus-wide for intro CS homeworks, reaching 500+ students per year
- Deployed user-friendly website to parse and display problem-wise Feedbot results to students
- **Lead TA** for a lab section with 25+ students and held weekly office hours to reinforce lecture concepts
- Co-authored a research report: "*Feedbot: Formative Design Feedback on Programming Assignments*"

Synth Research Project, Northeastern University *Jan – June 2024*

- Developed an LLM-based Program Synthesis tool using Python and OpenAI APIs
- Incorporated Doctests and Property-based tests in Python as specifications for synthesis
- Developed efficient caching and dynamic re-querying mechanism in case of failed synthesis.
- Performed user testing for tool with **100+ students** in lab sections for a CS course
- Authored a (draft) research paper: "*Programming with non-algorithmic specifications*"

Interests: Formula 1, Programming languages, Cricket, Dogs